

Examiner reports

Q1.

This question also proved to be a good source of marks for a majority of candidates. Normally part (a) was answered correctly but there were also signs of poor examination technique. Candidates who just wrote down the y -coordinate of P as '3' scored no marks but those who wrote ' $3(2^0 + 1) = 3$ ' scored 1 of the 2 marks available. Responses to the trapezium rule question were mixed with many candidates scoring full marks but a greater number of candidates failing to pick up even the method mark. The two main method errors involved either the use of seven ordinates rather than the required four or the use of $x = 0, 1, 2, 3$ only.

In part (c)(i) a significant number of candidates were unable to gain the mark because they wrote 3×2^x as 6^x . Many candidates gained the 2 method marks for using logarithmic rules but a significant minority failed to round 2.5849 correctly to three significant figures. The wrong answer 2.59 was seen more than expected.

Q2.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14 \ln\left(\frac{5}{12}\right)$ or $-14 \ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression $\frac{1}{14}$ for x , although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince the examiner that they had found the requested result, with much abuse of the $\ln$ function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x , and some very neat solutions were seen this way, although those who did not consider limits or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for

applying the result from part (c)(i), although some did say that $\frac{1}{14}(7) = 2$.

Q3.

A majority of candidates indicated that the method for solving equations of the form $a^x = b$ was well known. The most common errors are illustrated by

(a) $\log x^{0.8} = \log 0.05$

(b) (i) $x = \frac{\log 0.8}{\log 0.05}$

- (ii) answers 13.4 or 13.43 Part (b), which tested geometric series, although answered much better than the corresponding question in summer 2005, still indicates a general weakness in this area of the specification and the associated algebraic manipulation. It was not uncommon to see candidates using a specific value of a , either 1 or 20 or even 100, to solve the equation

$$\frac{a}{1-r} = 5a$$

In the final part a significant proportion of the candidature used the formula for the sum to n terms rather than the formula for the n th term. Some better candidates obtained $n = 14.5\dots$, but failed to realise that n was an integer.

Q4.

- (a) The correct value of a was often seen, but $a = -9$ was a common incorrect answer. The majority of candidates gained the method mark in finding b , but a number of candidates then continued with incorrect manipulation of the exponential function.
- (b) The majority of candidates answered this correctly.

Q5.

- (a) Reasonable attempts were made by most candidates. The main mistake was finding the derivative of x^{-2} which was often seen as $-2x^{-1}$. This lost marks since candidates then 'found' the answer given by incorrect means.
- (b) There were many fully correct answers but many candidates chose incorrect functions for u and $\frac{dv}{dx}$, with the obvious implications.
- (c) (i) This was reasonably well answered, the main error seen being $1 - 21nx = x^3$.
- (ii) The majority of candidates obtained the method mark following through from their answer to part (b). Few candidates completed this question correctly.

Q6.

This question was answered very well with most candidates gaining nearly full marks. In part (a) many wrote £80 instead of just 80 but this was accepted.

Part (b) was mostly done accurately with many getting to $k = \sqrt[50]{62.5}$ which was accepted

as *showing that*. Those who used logarithms lost a mark if they didn't show how $\log 62.5 = 561 \log k$: led to the given value of k .

Virtually all candidates got part (c) correct, with both the approximate answer 200707 and the more exact 200648 being seen, often with several decimal places attached, but this was of no matter as candidates had demonstrated a value greater than £200,000. In part (c)(ii) there were occasional errors although most candidates used logarithms to derive 124.7 correctly whilst a few used trial and improvement to deduce 125. However, many lost the last mark in incorrect interpretation of the year, or through not actually giving a year. The year 2025 was the common wrong answer.

Q7.

This question proved to be a good source of marks for many candidates. Most candidates answered both parts of (a) correctly although the usual errors, $3^\circ = 0$ or $3^\circ = 3$ and $3^x = 27 \Rightarrow x = \sqrt[3]{27}$ were seen. The trapezium rule, required in part (b), was generally well understood although some candidates are still mixing up 'ordinates' and 'strips' or failing to use sufficient brackets correctly. Although many candidates scored well in parts (c) there were others in (i) who used the relevant law of logarithms incorrectly or others who failed to indicate that logarithms had been used at all. In (ii) not all candidates realised that a simple substitution of 13 for 3^x in $k = 27 - 3^x$ with evaluation was all that was required. Descriptions of transformations continue to pose a problem for candidates. Many candidates gained partial credit for their sketches but both marks were rarely awarded.

Q8.

This question proved to be a good source of marks for many of the candidates. Its structure enabled candidates to progress even if they were unsuccessful with some of the parts.

Part (a)(i) Reasonably well done. Errors were with the derivative of e^{2x} . Common errors were e^{2x} and $2x e^{2x}$. A number of candidates added a '+ c' when they were differentiating.

Part (a)(ii). Usually correct, with errors, if seen, similar to those in the previous part.

Part (b)(i) The majority of candidates answered this correctly.

Part (b)(ii) Very well answered by the majority of candidates.

Part (b)(iii) Again very well answered with many totally correct responses. The major error apart from those who used the incorrect equation $e^{2x} - 5 e^{2x} + 6$ was the evaluation of e^{21n3} which was often seen as 6 rather than 9. Possibly the evaluation of e^{21n2} as 4 was fortuitous for some candidates.

Part (iv) Similar to part (iii) with similar errors, the major one being the evaluation of $4e^{21n3}$ as $4 \times 6 = 24$. This resulted in an answer of -6 and hence loss of accuracy marks.

Q9.

Most candidates struggled with part (a) of this question, indicating inexperience, or non understanding, of the formulation of a differential equation from a context situation.

Part (a) Despite the question explicitly requesting a differential equation for $\frac{dx}{dt}$ many candidates just wrote down an expression for x .

A commonly seen differential equation was $\frac{dx}{dt} = kx$ which candidates then proceeded to solve, despite the question stating that a solution of the differential equation was not required. Of those who could translate the given information into correct symbols, a symbol t for time, often found its way into their expression, and then often in an exponential form. Many attempts using the exponential function were seen, suggesting many candidates associating anything to do with differential equations with exponential forms. Those candidates who did give a correct differential equation in part (i) usually found a correct value for the constant of proportionality k . However, there were also many attempts from a partially correct part (i) where the given values of 1000 and 200 were confused, the latter often being taken as a value of time. Other candidates put forward the incorrect argument that the population would be equal to 1200 at time $t = 1$.

Part (b)(i) Virtually all candidates scored at least one mark here; the common error being to round off 5.545 to 5.6 or not to round it at all.

Part (b)(ii) Many candidates solved this equation correctly, although it was not always clear just how they had done it. Most candidates used the technique as per the mark

scheme, although some worked successfully from $e^{30} = \left(\frac{4x}{5000 - x}\right)^4$. Of those who went awry, the error usually involved an incorrect manipulation of the logarithm expression, including ignoring the $\ln$ altogether, or stating $4\ln 4 = \ln 16$. Some started correctly with $7.5 = \ln 4x - \ln (5000 - x)$ but obtained $e^{75} = 4x - (5000 - x)$. Others made relatively simple errors in signs or coefficients in their manipulation towards a solution.

Q10.

Many candidates incorrectly evaluated $(3^0 + 1)$ as 1 in part (a) although surprisingly they frequently recovered and used the correct answer, 2, in part (b). In general, the trapezium rule was well understood, although it was relatively common to find an arithmetical slip in evaluating $(3^x + 1)$; the '1' was not added in one of the terms. Candidates should have given the final answer to the degree of accuracy specified in this numerical integration question. Some of the weaker candidates just integrated $(3^x + 1)$ directly as $(3^{x+1} + x)$. The use of a diagram to show that the approximation is an overestimate was well understood, although some used rectangles rather than the required trapeziums and others had the sloping sides of their trapeziums in the wrong positions, that is, below the curve. Part (c) was generally well answered although some weaker candidates tried to take logarithms of each term rather than to rearrange the equation into the form ' $3^x = 4$ ' as a first step. Those candidates who used trial and improvement usually failed to produce an acceptable solution. In part (d), a common wrong expression for $f(x)$ was " $-3^x + 1$ ", although a significant minority did obtain the correct answer " $3^{-x} + 1$ ".

Q11.

Part (a) tested the relationship of e^x and $\ln x$. This part was well answered. Part (b) required algebraic manipulation using e^{-x} which discriminated between candidates. The final part was accessible to candidates who failed on the previous part. Candidates were required to give exact answers and not decimal equivalents.

Q12.

Many candidates gained full marks on this question presenting clear, good quality answers.

Part (b)(i) Most candidates knew they were required to separate and integrate and attempted to do so, sometimes with poor notation and incorrect integrals but their intention was clear. Most candidates also included a constant of integration and attempted to find it using the given initial conditions. Some neglected to express t in terms of x with some going straight to the result given in part (b)(ii).

Part (b)(ii) This was often done well but the common error was in taking exponentials, with

$e^{\frac{t}{40}} = e^{\ln 70} + e^{\ln(x-15)}$ being fairly common.



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